Proceedings of the 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS-2025)
IEEE Xplore Part Number: CFP25AZ5-ART; ISBN: 979-8-3315-1250-7

2025 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS) | 979-8-3315-1250-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSCDS65426.2025.11167845

CyberDome: Cloud based Intrusion Detection and Mitigation

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***Abstract*—The current Internet of Things (IoT) security systems are unable to keep up with the pace of increasing things and the complexity of their structure, resulting in need for scalable, robust and contemporary systems capable of real-time security. The proposed system conceives a combination of cloud-based Intrusion Detection System (IDS) along with a mitigation module. The cloud based IDS comprises of machine learning model (XGBoost) with F1-scores ranging from 0.995 to 1.0, capable of classifying the network traffic logs and invoking the mitigation module on anomaly detection. Mitigation module based on Software Defined Networks (SDN) can mitigate threats using open flow rules taking up tasks like dynamic flow control, network segmentation and device quarantine on demand by IDS. The mitigation module is also a cloud-based component, with the term SDN as a Service, which possess the capability to manage multiple large complex networks while maintaining the network agility, security and scalability under heavy traffic loads. Machine learning model deployed in the cloud, XGBoost is a lightweight model adept to generalizing to unseen data cleverly avoiding overfitting by Incorporating regularization techniques. When number of networks are scaled, among the plausible solutions, the cloud based solution proves to be cost effective while maintaining a prominent performance compared to edge computing or TinyML-based solutions with reduced management overhead along with easy and centralized maintenance.**

***Index Terms*—Cloud, Machine Learning, SDN, Mitigation module, Intrusion detection, Network, Network traffic, IoT devices, Autonomous security system, XGBoost, dynamic flow control.**

I. Introduction

The enhancements and rapid improvements in the field of Internet of Things (IoT) have enabled professionals from various industries to be able to build highly dynamic large-scale networks of 'things'. These things can belong to a wide range of use cases such as environmental sensors and home appliances to industrial actuators that are constantly exchanging sensitive data in real time through various gateways and mediums [11]. These networks and deployments have their advantages in performing invaluable actions and this pervasive connectivity enables a plethora of new applications and websites, but it definitely comes with its cons such as expanding the attack surface. Under these circumstances, the traditional and legacy-based solutions that are currently being utilized, also known as perimeter-based security solutions prove ineffective. These solutions cannot adapt quickly enough to these evolving threats, nor can they provide a robust and scalable mechanism to monitor and keep track of thousands of heterogeneous endpoints without overwhelming devices and network resources.

To address these problems, the proposed system, CyberDome, is a robust and centralized solution, a cloud-native IoT security monitoring system that integrates a high-fidelity, machine learning-driven Intrusion Detection Module, and a Software-Defined Networking (SDN)-based Mitigation Module [12]. The entire system has quite an intuitive design with the network edge, handling the functions like raw packet capture being parsed and converted to CSV using industry wide used inbuilt python library PyShark, then preprocessed by a lightweight scaler, this largely offloads the computational overhead and preserves data privacy. The core component of the system is the machine-learning driven module deployed in the cloud used for anomaly detection, an XGBoost classifier is trained to distinguish among eleven traffic classes (one benign plus ten distinct attack types such as DoS, Mirai botnet variants, ARP spoofing, port/OS scanning and telnet brute-force), achieves astounding

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evaluation metric scores. The model achieved F1-scores between 0.995 and 1.000. Upon detecting any anomaly, the IDS module automatically invokes the SDN controller, which follows OpenFlow rules to perform multiple different actions such as dynamic flow, rate throttling and virtual segmentation (via proxy routers) to full device quarantine.

The proposed system's cloud architecture unifies global visibility, user management and real-time network visualization. The flow of the systems is handled by distinct FastAPI-driven dashboards, which includes registered clients dashboard and admin dashboard [13]. There is an underlying PostgreSQL database to perform various data representation tasks, ranging from representing live topological graphs, per-device status and recent traffic logs. New networks can be added by the admin which maintain persistent state across sessions, the mitigation module enforces device level mitigation states which dynamically changes with tunable timers, eg., 10 s flow control then a 15 s segmentation followed by 15 s quarantine, before restoring to default state and operation [14]. By utilizing enhancements in the aforementioned fields edge preprocessing, centralised ML inference and SDN based mitigation, CyberDome delivers a robust, scalable, low- latency framework that secures IoT ecosystems under heavy loads from multi-tenant systems while minimizing overhead and false calls.

II. Literature Review

The surge in IoT and its usage has left us with a number of new challenges in using it effectively. These above mentioned challenges primarily include network security, particularly in detecting and dealing with these threats in highly heterogeneous and distributed environment. There are various studies delving into this topic to explore the most effective and feasible methodology to build a system to counter these emerging threats.

[1] evaluated multiple machine learning models, especially supervised learning algorithms including K-Nearest Neighbors (KNN), Naive Bayes (NB) and convolutional Neural Networks (CNN) on the Bot-IoT dataset. The authors have come to a conclusion that CNN-based models outperformed traditional machine learning classification models with over 99.9% accuracy in detecting threats. Their work however lacked considerations regarding deployment and the mitigation module backed up with SDN integration.

[2] proposed a very unique solution utilizing the concept of device fingerprinting that uses traffic-based features to help us identify and also classify IoT devices. The method put out promising results with very high accuracy (90-97%) for 27 device types, but it lacks cloud-based hosting or mitigation methods for anomaly detection and handling.

In a related study, [3] has also given and specified the key principles for performing secure operations on IoT systems, such as blockchain for immutable audit logs and fog computing for decentralized data processing. They stressed on the importance of future proofing these systems by building strong hybrid defense mechanisms such as cryptographic protocols, federated ML models and zero-trust architectures.

[4] has built a very secure end-to-end simulation of IoT network architecture using NetSim. The proposed system is a combination of lightweight edge encryption along with a cloud-hosted ML-based intrusion detection module. However, the explored system lacks the presence of dynamic SDN-based response mechanism.

[5] introduced a hybrid model for detecting threats using the concept of Principal Component Analysis (PCA) for dimensionality reduction making it easier for the model to train and simplifying the workings of the model, along with XGBoost as classification model. The model achieved over 98% accuracy on the well accepted UNSW-NB15 dataset, which proved itself for resource-constrained environments. The paper however lacks specifications in the fields of real-time mitigation or system deployment in IoT networks.

[6] explored QoS-based SDN frameworks combined with ML-based intrusion detection, this does an excellent job in dynamically adjusting OpenFlow rules based on ML model verdicts, this helps have a better record in protecting against DDoS attacks. This study has laid the necessary groundwork for cloud-managed SDN security systems.

[7] gives the most comprehensive overview of techniques used in detecting threats and anomalies in cloud computing, at the same time giving summary on various machine learning models including models like SVM, K-Means, and Autoencoders. This analysis has put lot of stress on the trade-offs between signature-based and anomaly-based detection in highly elastic cloud environments, while shedding light on the need for continuous model retraining and adaptive pipelines.

[8] has put forward an ensemble model combining Isolation Forest and Local Outlier Factor (LOF) to detect anomalies in wireless networks. This does improve the detection rates, but the paper lacks deployment based scalability or integration with real-time mitigation modules.

[9] has a very unique approach with layered-based security framework for IoT networks utilizing secure boot, blockchain-backed logging, hardware security modules (HSM's) and ML-based intrusion detection. This is a very theoretical concept with an operational complexity and high cost for real-time deployment in large-scale IoT environments.

[10] has built an Autoencoder-based anomaly detection

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which combines with Prometheus and is hosted in cloud- native environments using Kubernetes. The agent is also retrained from time-to-time to handle concept drift and work in both loop and one-shot modes.

From reviewed literature, it shows that while existing systems have very high accuracy, only few systems and models provide complete end-to-end solutions that integrate real-time anomaly detection along with cloud-native deployment and SDN-based dynamic mitigation. The noticed gap is motivation behind the development of CyberDome, which is a unified, scalable and actionable framework that performs operations of IoT anomaly detection and mitigation in a cloud and SDN hybrid environment.

III. Design and Methodology

The proposed system sets the ground for the mitigation module using the features of SDN when an intrusion is detected in the intrusion detection module. The machine learning model used in this proposed work is XGBoost which is lightweight while boasting higher accuracy, making it perfect for this kind of system where performance and accuracy are crucial. The machine learning model is trained and deployed in the cloud. The proposed system uses the SDN in the cloud itself like SDN as a Service, where the SDN can mitigate the intrusion detected by the IDS. The intrusion caused by the attacker can be mitigated using the SDN's OpenFlow rules where the logical view can be changed without the interference with the physical devices or the need to change the actual topology of the devices to segment a network, apply dynamic flow control and quarantine devices; which can help to mitigate the threats. Figure 1 shows the architecture of the proposed system.

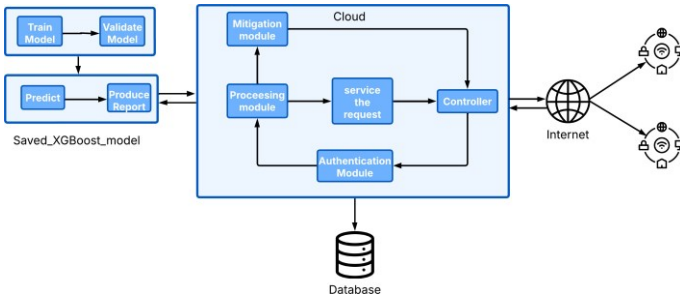


Fig. 1. System Architecture

A. Overall System Architecture

Our proposed system comprises the following.

- ML pipeline: This is the basic structure that handles preprocessing and training of the model for the deployment of the model in the cloud environment. Handles data ingestion, preprocessing, feature engineering and then follows model training and evaluation using various standard metrics.
- IoT Security Service: The IoT security service is provided by the mitigation module with decisions based on the classification provided by the intrusion detection system. The IoT security service uses SDN to provide security features to network devices. The security system also has Network IDing and Device IDing where the cloud provides them with unique network ID and device ID based on the network ID given by the cloud to prevent shadow IoT devices.
- Edge Computing: The data that is being sent to the cloud is preprocessed using the edge devices to reduce the overhead of preprocessing in the cloud where it can be sent to inference directly for classification.

There are pipelines created to invoke the mitigation module when the classification is an anomaly and pipelines to provide interface for the network admins to make the network visualization for nontechnical persons to understand the state of the network and providing the tools to unquarantine the devices if needed.

B. Network initialization and registering

when the network is first setup to connect to the cloud the network is provided with the user ID which belongs to the network admin that is used to register to that particular user in the cloud. When the network registration is started a unique network ID is generated by the cloud for the network which will be stored by the network so that even when the network restarts it will have a memory of its registration. Each network ID is unique and device ID can be generated using this network ID fingerprinting each device of the network. The network ID is stored in a database on the cloud side to the user it is registered to.

C. Feature Engineering

All the aforementioned scripts extract high-level attributes from the raw packets captured, these include attributes like per-flow, byte counts, packet rates and much more. As per the standard practice categorical fields are one-hot encoded and the final vector $f \in R^d$ feeds straight into the model that is being trained XGBoost without further runtime preprocessing. The feature engineering is done so that it can handle different type of headers such as the UDP, TCP etc.

D. Model Training

The model is trained by the train_model.py module which implements XGBoost classifier with a softmax objective for 11-class detection. The key hyperparameters included a tree depth of 6 and a learning rate of 0.1. The models are then validated on some of the standard evaluation metrics, merror, and weighted accuracy, and the better model is saved. The best model obtained from the training was then deployed into the cloud server for inference.

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E. Inference and Real-time service

At runtime:

- Packet reception takes place, where the client systems forward POST JSON vectors to and API endpoint, the server then surrounds them in an DMatrix belonging to the XGBoost model and then the model infers the data to provide an accurate class prediction along with the meta data such as timestamps and time logs. To improve performance and keep the workflow clean, the system has been incorporated with an in-memory deque data structure that retains the most recent 25 entries for each registered network.
- IoT gateways periodically POST their device graph and status which helps to visualize networks in the user interface. The interfaces being mentioned are as follows:
  - Client Dashboard: Displays the network topology of the user, along with the network logs for the last 25 classification requests. The status per-device is also specified with the states being active, under flow control, segmented, quarantined.
  - Admin Console: Displays registered users along with registered networks and network counts.

F. On-Demand Mitigation via SDN

As mentioned above upon detection of non-normal or anomalous classification, background tasks call a three-staged mitigation module which works on the backdrop of the SDN controller. The attack severity can be taken into account to skip the three step process to mitigate and directly quarantine a device to prevent the lateral movement of the intrusion in the network devices.

The main aim of the proposed system stands to mitigate the threats and make some centralized updates to the security system without the need of user to interfere making it more easier and accessible. The cloud platform makes it better with auto scaling according to the needs of the user and expanding networks and devices.

IV. Analysis of Proposed Work

The CyberDome is made to handle multiple networks with multiple devices on board. The machine learning model in the proposed work was analyzed using the merror metric in the XGBoost which is used to evaluate multi-class classification models. The merror at the end of training the XGBoost model is 2.462858722214e-05. The accuracy can be given as $accuracy = 1 - merror$ the accuracy during the training is given as 99.9975% and the accuracy in the test set is 99.9945%. Figure 2 shows the accuracy of the model during the training of 100 boosting rounds. The accuracy is sharply increased up to 20 rounds, the improvement is flattened from 20 to 40 rounds then the accuracy of the model is slowly improved up until the 100 rounds reaching the final accuracy of 99.99%. Both the accuracies in the training set and the test set are of 99.99% which shows that there is no overfitting of the model.

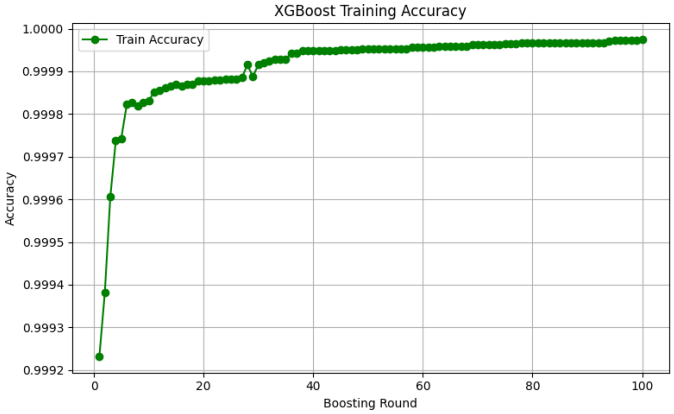


Fig. 2. Accuracy during training

The loss graph in the figure 3 shows the confidence of the model in its predictions. The low value of loss means that the model has high confidence in its predictions. An exponential decrease in the graph can be noticed which indicates that the model's accuracy is improved an it makes more accurate predictions over the training time.

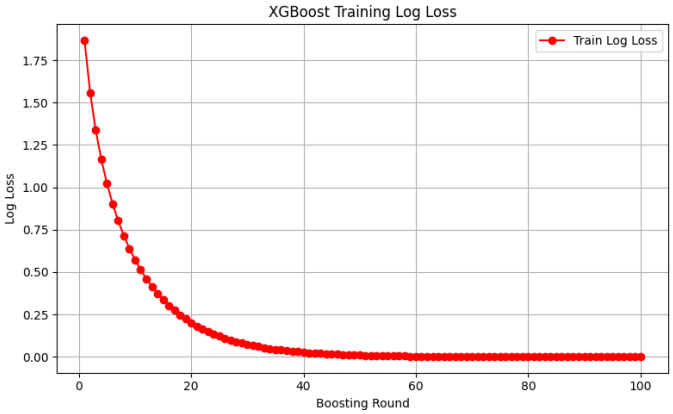


Fig. 3. Log Loss during training

Table I compares five model's F1 Macro avg and the avg inference time per datapoint in seconds. The XGBoost and the random forest models have the least avg inference time per data point making them suitable for intrusion detection.

TABLE I
Model Comparison: F1 Score and Average Inference Time per Data Point

Model	F1 Macro Avg	Avg. Inference Time / Point (s)
CNN	0.999	4.6025e-05
KNN	0.997	1.7755e-03
Random Forest	0.999	6.5626e-06
SVM	0.998	1.7521e-04
XGBoost	0.999	8.7668e-06

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The table I consists of the F1 macro avg of the models which represents the performance of the multi classification model. The KNN is the least favored algorithm due its low performance in the classification, the SVM algorithm performs better than the KNN but falls behind the CNN, random forest and the XGBoost algorithms which perform better in classifying the attack packets with varying inference average inference times with the random forest having the best average inference time among them followed by the XGBoost.

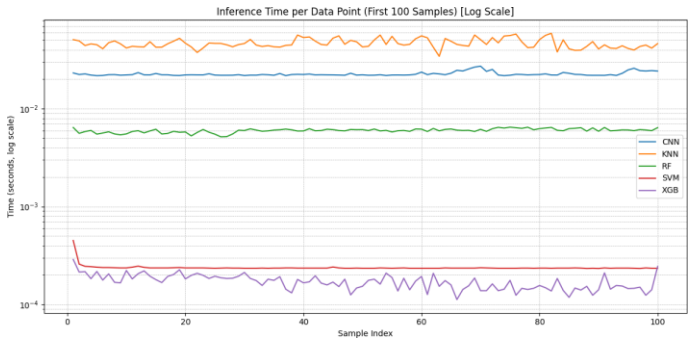


Fig. 4. Inference Time comparison of different models

In the figure 4 five machine learning are compared based on their inference time per data point. The graph is based on the first 100 samples of the test dataset. The XGBoost model undercuts the other models time in almost every data point for the first 100 samples in the test dataset. The average inference time can be increased with some rare data points where the inference time is high so the most possible inference time is taken under consideration in which the XGBoost model is the best one for real world scenario with the least inference time among the compared models.

V. Results

In the implemented proposed system, the accuracy of the XGBoost model is ranging from 0.995 to 1.000 with the Host Discovery have the least accuracy of 0.995 and with Mirai ACK flooding having the highest accuracy of 1.000 with the decimals rounded of to 3 places

The model trained for 39.601158999999996 seconds and the average inference time for preprocessed data point is 8.737001685676134e-06 seconds. This machine learning model is pipelined to receive the preprocessed data from the network and use it for classification of packets based on the label mapping. The classification of the packet is represented with the device from which the classified packet emanated from.

The mitigation module is employing the three-staged mitigation for infected devices the first stage is dynamic flow control. The device is put in network segmentation state where the number of packets emanating from the device is reduced. This stage will last for limited time while still inferring the network packets of the controlled devices, after some time if there are no attack packets detected from the device then the device is put back into its original state where it retains its full functionality. Figure 5 shows flow control in a simulated network.

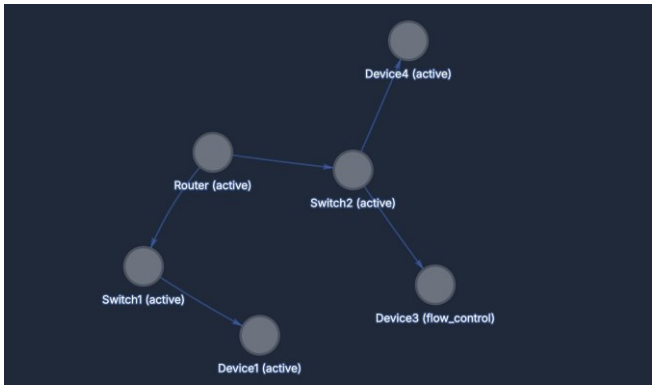


Fig. 5. Flow Control

If there are more attack packets from the flow controlled device then the device is put in network segmentation state which is achieved by creating a virtual router and connecting the infected device to it, this makes sure that the device is still functioning by making sure it is connected to the internet. Figure 6 shows the segmentation of the simulated network.

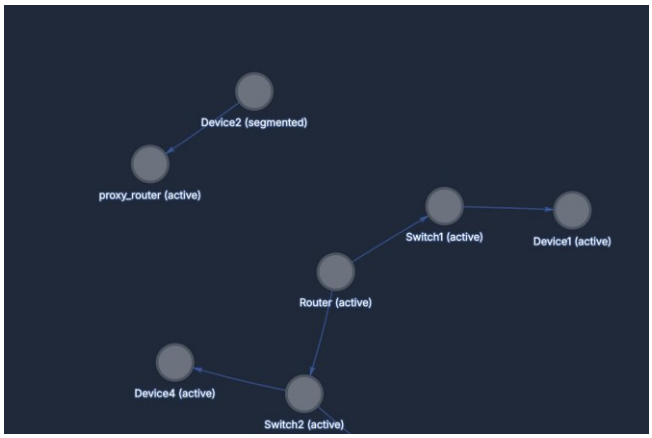


Fig. 6. Network Segmentation

Even after the device is segmented from the network if there are more attack packets from the device, the device is quarantined; this state can also be directly achieved when the attack classification is more severe, such as Mirai, MITM, and DOS attacks. Figure 7 gives the device quarantine in a simulated network.

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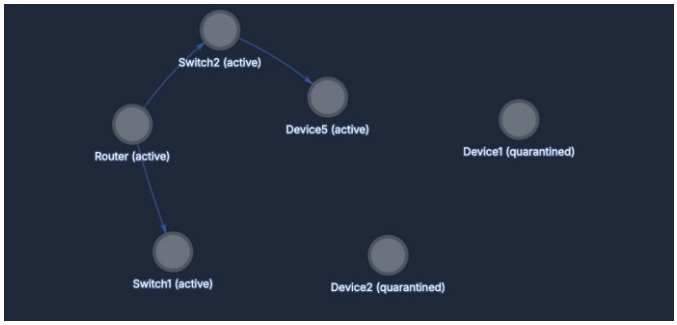


Fig. 7. Device Quarantine

CONCLUSION

CyberDome, is a cloud-based security system incorporating machine learning models for intrusion detection with SDN for mitigation tasks, demonstrates a robust solution for intrusion detection and mitigation for heterogeneous networks. Utilizing current technologies such as XGBoost for anomaly detection and SDN for dynamic mitigation, the system solves multiple functional and nonfunctional requirements like scalability, adaptability, and real-time responsiveness challenges which are innate to modern IoT environments. Evaluating the architecture against multiple attack types and vectors revealed high accuracy and low false positive rates. Combining edge-level preprocessing further distributes computational load, while preserving data privacy. The intrusion detection can be vastly enhanced by employing advanced feature extraction like deep autoencoders to distill relevant IoT traffic patterns and reduce the dimensionality in the input. Further improvements can focus on improving the model robustness by combining XGBoost with ensemble, reinforcement learning methods and federated cloud-edge learning to counter advanced high level threats and also the zero-day vulnerabilities. IP traceback using the SDN can be used to pinpoint attack sources and enforce OpenFlow rules at the origin enhancing the mitigation module. Real-time detection will benefit from lightweight ML models perfectly optimized for edge deployment. The newly developed zero-trust architectures, along with hardware strengthening would reinforce endpoint integrity. Additionally, using 5G network slicing for secured channel isolation and post-quantum cryptographic schemes would safeguard the system against threats that are evolving rapidly. Finally, compliance with global benchmarks and guidelines will ensure greater long-term relevance of the proposed framework in the IoT security system.

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